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RAMAPURAM CHENNAI

4th International Conference on

Challenges and Opportunities in Renewable Energy, Smart Systems & E-Mobility (ICCORSE 2026)

12th May 2026 (Hybrid Mode)

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ADAPTIVE SWARM-ACTUATED NET SYSTEM FOR SPACE DEBRIS CAPTURE: INTEGRATING DISTRIBUTED ROBOTICS WITH FLEXIBLE CAPTURE MECHANISMS

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ABSTRACT

"The increasing density of orbital debris poses a critical threat to satellite operations and long-term space sustainability, potentially leading to cascading collision scenarios. Conventional debris removal approaches, including single-satellite net systems and rigid capture mechanisms, suffer from limited adaptability and high failure risk when interacting with non-cooperative, tumbling targets. This paper proposes an Adaptive Swarm-Actuated Net System (ASANS) that integrates distributed swarm robotics with a flexible, multi-point actuated net for efficient debris capture. The proposed system utilizes a coordinated network of small satellites to dynamically deploy, shape, and control a deformable net structure in real time. By leveraging decentralized control algorithms, adaptive tensioning strategies, and multi-agent sensing, the system enhances capture reliability and robustness in uncertain orbital conditions. A system-level architecture, operational workflow, and control strategy are presented, along with an analysis of key engineering challenges including orbital dynamics, communication constraints, and nonlinear net behaviour. Simulation-based validation strategies and feasibility assessments are discussed. The proposed approach demonstrates strong potential as a scalable and fault-tolerant solution for next-generation active debris removal missions."

ISBN: 978-81-990189-7-6